

WASP-135b

R-0830

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_260623 · created 2026-07-28T17:08:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461214.872 +/- 0.016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.049
Transit depth (Rp/Rs)^2	1.99 +/- 1.39 [%]
Orbital Inclination (inc)	86.95 +/- 2.95
Airmass coefficient 1 (a1)	1.285 +/- 0.032
Airmass coefficient 2 (a2)	-0.187 +/- 0.038
Scatter in the residuals of the lightcurve fit is	2.52 %
Best Comparison Star	None
Optimal Aperture	3.69
Optimal Annulus	16.31
Transit Duration (day)	0.063 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.049 vs 0.139 (deviation 0.04σ)
Duration fit vs published	0.063 d vs 0.0687 d (deviation 0.35σ)
Mid-time O–C	-8.6 min
Depth SNR	2.8
Residual scatter	2.52 %

Light curve

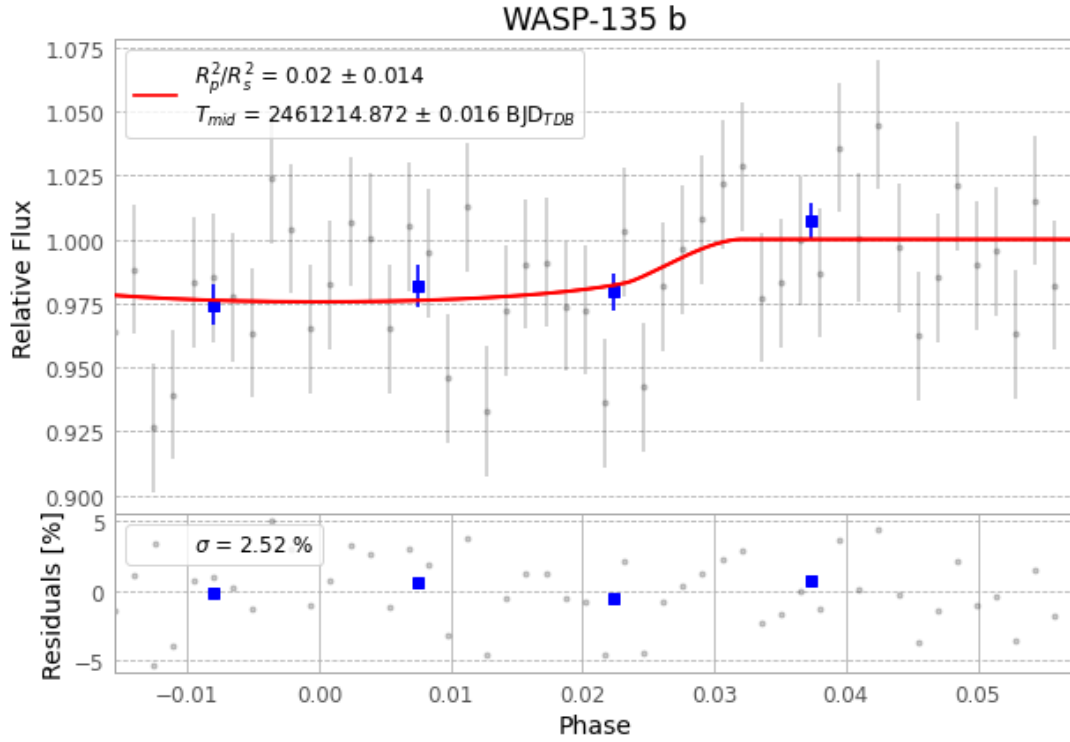


Figure 1 — FinalLightCurve_WASP-135 b_2026-06-23.png (SHA-256 01e8a0bc11d59205...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [180, 177], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9b1e387b9836d6f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

50 FITS frames (33 MB), WASP-135260623081816.FITS → WASP-135260623104515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260623.log (907cfaa06da1...), FinalLightCurve_WASP-135 b_2026-06-23.png (01e8a0bc11d5...), FinalLightCurve_WASP-135 b_2026-06-23.csv (ed388bfe3eff...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 17:08Z.